Figure 3.5.1.3. Now by moving the gear shift fork on the left as shown in Figure the dog clutch is allowed to mesh with the dog clutch in the clutch shaft.

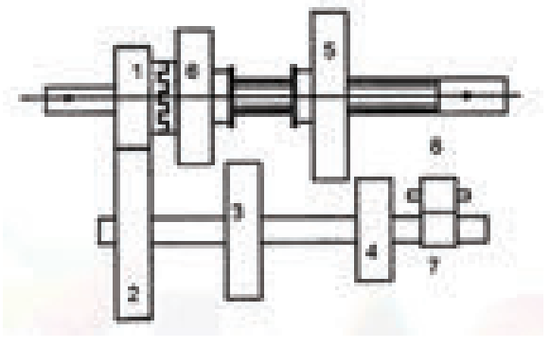


Figure 3.5.1.3 Third gear power transmission

Now the output shaft rotates at the speed same as the clutch (main) shaft. This is the maximum speed and the gear ratio in this condition is 1:1. The rotational power is now directly transmitted from the clutch shaft to the output shaft.

e) Reverse gear

For reverse gear, the gear shift fork is moved towards the right direction as shown in the Figure 3.5.1.4. Hence the gear 5 in the output shaft is in mesh with the gear 8 in the lay shaft which is the idler gear. Now the power transmission is from gear 1-2-7-8-5 to the output shaft.

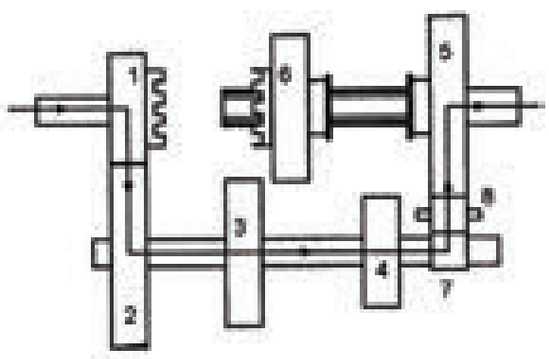


Figure 3.5.1.4 Reverse gear power transmission

Advantages of sliding mesh gearbox

1. Simple in construction
2. Simple in maintenance
3. Suitable for heavy vehicles
4. Low cost
5. Higher mechanical efficiency
6. Production cost is low

Disadvantages

1. Gear changing is difficult
2. Noisy in operation
3. Experience is required for changing gear
4. The edges of teeth wear rapidly.

3.5.2 Constant mesh gear box

In constant mesh gear box the teeth in the main shaft and lay shaft are always in constant mesh with each other. Hence it is called as constant mesh gear box. Helical gears are used in this type of gear box. The gear arrangements in the constant mesh gear box can be seen in Figure 3.5.2.

Construction

In the constant mesh gear box the gears 6, 7 and 8 in the output shaft are always in mesh with the gears 5, 4 and 3 in the lay shaft. In the clutch shaft and in the lay shaft there are gears 1 and 2 respectively which are always in mesh with each other. The gears in the main shaft are supported by bushes. Gear 5 in the lay shaft is in mesh with an idler gear. After the clutch gear in the clutch shaft and before the reverse gear in the main shaft there are dog clutches (D_2 and D_1 respectively) present. These dog clutches

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